Notes on

Multivariate Calculus

Lecturers

Riddhick Birbonshi

Sanjib Kumar Pal

Notes typed by

Sayan Das

Fifth Semester 2025

Jadavpur University

This text consists of notes on the course Multivariate Calculus, lectured by Professor Riddhick Birbonshi (RB) and Professor Sanjib Kumar Pal (SKP) at Jadavpur University in the 5th semester 2025.

Some changes and some additions have been made by me. To distinguish them from the lecture's actual contents, they are labelled with an asterisk. So any *Lemma** or *Remark** or *Proof** that the reader might come across are wholly my responsibility.

Please report errors, typos etc. to this email: dassayan0013@gmail.com.

Contents

List of Lectures

Course organisation. — The lectures will take place on Tuesdays from from 14:30 to 16:30 and on Wednesdays from 14:30 to 16:30, in the UG-III classroom on the ground floor of Ramanujan Bhavan.

The prerequisites to this lecture are a course in linear algebra covering the content of these first and fourth semester courses at JU and a series of courses in analyis of a single real variable, akin to the following courses: Real Analysis, Theory of Real Functions and Riemann Integration and Series of Functions. A knowledge of metric spaces and basic differential geometry of curves and surfaces is also assumed, at a level akin to the fourth semester course Metric Space and Differential Geometry. For this course, we will follow the chapters in Rudin ([Rud53]) and Apostol ([Apo74]) which deal with functions of several variables. I personally really like the two-volume text by Duistermaat and Kolk ([DK04a] and [DK04b]).

The syllabus of this course is available here.

CHAPTER 1.

Differentiation in $\mathbb{R}^n$

1

1.1. Limits and continuity

LECTURE 1
6^{th} Aug, 2025

1.1.1. $\mathbf{R}^n$ as a normed linear space. — *What is* $\mathbb{R}^n$ *?* It is the set of all n-tuples with components in $\mathbb{R}$,

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}.$$

For, $x, y \in \mathbb{R}^n : x = (x_1, x_2, \dots, x_n), y = (y_1, y_2, \dots, y_n)$ we say that $x = y$ iff they are componentwise equal, i.e.

$$x = y \iff x_i = y_i \text{ for } i = 1, 2, \dots, n.$$

For $\lambda \in \mathbb{R},\ x, y \in \mathbb{R}^n : x = (x_1, x_2, \dots, x_n), y = (y_1, y_2, \dots, y_n)$ we define the following operations componentwise

$$\lambda x = (\lambda x_1, \lambda x_2, \dots, \lambda x_n)$$
$$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n).$$

So $\mathbb{R}^n$ is an $\mathbb{R}$-vector space. $x - y = x + (-y)$ and the zero vector is the element $0 \in \mathbb{R}^n$ with all components zero i.e. $0 = (0, 0, \dots, 0)$. Then clearly this is the additive identity in $\mathbb{R}^n$.

We also have the (Euclidean) inner product or dot product

$$\boxed{x \cdot y} = \sum_{k=1}^{n} x_k y_k.$$

The length of a vector is defined using the (Euclidean) norm,

$$\|x\| = (x \cdot x)^{\frac{1}{2}}$$

which is the distance from 0 to x. The distance between any two points $x, y \in \mathbb{R}^n$ is $\|x - y\|$.

In particular, for $n = 2$, the Euclidean distance between two points is given by the familiar distance formula

$$\|x - y\| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

Moreover, if we take $d(x, y) = \|x - y\| : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ then $(\mathbb{R}^n, d)$ is a metric space. In fact, it is a *normed linear space* or *normed vector space*.

It is also complete, which makes it a *Banach space*.

1.1.2. Definition (Normed linear space)**.** — Suppose V is an $\mathbb{F}$-vector space, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The **norm** $\|\cdot\|$ on V is defined by

$$\|\cdot\| : V \to \mathbb{R}^+ \cup \{0\}$$

such that it satisfies the following properties:

(a) $\|x\| \geqslant 0$ for all $x \in V$.

(b) $\|\alpha x\| = |\alpha|\|x\|$ for all $\alpha \in \mathbb{F}, x \in V$.

(c) $\|x+y\| \leqslant \|x\| + \|y\|$.

Then the vector space V endowed with the norm $\|\cdot\|$ is called a **normed linear space**.

In particular, $\mathbb{R}^n$ is a normed linear space with respect to the Euclidean norm. We also have the Cauchy-Schwarz inequality which holds in any normed linear space: in particular for $\mathbb{R}^n$, if $x, y \in \mathbb{R}^n$

$$\left|\sum_{k=1}^{n} x_k y_k\right| \leqslant \left(\sum_{k=1}^{n} x_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^{n} y_k^2\right)^{\frac{1}{2}}.$$

Every normed linear space is a metric space but the converse is, of course, not true.

1.1.3. Definition. — A function $f : \mathbb{R}^n \to \mathbb{R}^m$ (m and n not necessarily the same) is an m-tuple of functions $f_i : \mathbb{R}^n \to \mathbb{R}$, i.e.

$$f = (f_1,\ f_2,\ \dots, f_m) : f_i : \mathbb{R}^n \to \mathbb{R} \ \text{ for } i = 1, 2, \dots, m.$$

1.2. Continuity

1.2.1. Definition*. — Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $b \in \mathbb{R}^m$ and a be a limit point of A, then we say f has the **limit** b at a, denoted $\lim_{x\to a} f(x) = b$, if for every $\varepsilon > 0$ there exists $\delta > 0$ s.t.

$$x \in A \text{ and } \|x - a\| < \delta \implies \|f(x) - b\| < \varepsilon.$$

$(\mathbb{R}^n, d_{E_1})$ and $(\mathbb{R}^m, d_{E_2})$ are metric spaces where d_{E_1} and d_{E_2} are the respective Euclidean metrics on $\mathbb{R}^n$ and $\mathbb{R}^m$. Then equivalently

$$\lim_{x\to a} f(x) = b \iff (\forall \varepsilon > 0\ \exists \delta > 0 : A \cap d_{E_1}(x, a) < \delta \implies d_{E_2}(f(x), b) < \varepsilon.)$$

1.2.2. Definition. — We say $f : \mathbb{R}^n \to \mathbb{R}^m$ is **continuous** at some point $x_0 \in \mathbb{R}^n$ if for all $\varepsilon > 0$ there exists $\delta > 0$ s.t.

$$d_{E_1}(x, x_0) < \delta \implies d_{E_2}(f(x), f(x_0)) < \varepsilon.$$

1.2.3. Question (Problems). — (1) Let $f : \mathbb{R}^2 \to \mathbb{R}$ s.t.

$$f(x, y) = \frac{x^2 y^2}{x^2 + y^2},\ (x, y) \neq (0, 0).$$

Then show that

$$\lim_{(x,y)\to(0,0)} f(x, y) = 0.$$

(2) Show that

$$\lim_{(x,y)\to(0,0)} xy\frac{x^2 - y^2}{x^2 + y^2} = 0.$$

(3) Compute the limit

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2}.$$

LECTURE 2
12th Aug, 2025

1.2.4. Limit laws. —

1.3. Derivatives

Lecture 3
19^{th} Aug, 2025

1.3.1. Differentiability. —

CHAPTER 2.

Integration in $\mathbb{R}^n$

2

2.1. Curves in $\mathbb{R}^n$

LECTURE 4
5^{th} Aug, 2025

2.1.1. Directed curves. — As we'll be doing mathematics, not physics or engineering, we will study curves in not simply $\mathbb{R}^2$ or $\mathbb{R}^3$ but $\mathbb{R}^n$ from the get-go, and only appeal to the cases $n = 2, 3$ for concrete intuition. We will apply this to the following:

- Line integrals.
- Potential ($\vec{F} = \nabla\psi$).
- Stokes' Theorem.
- Geometry of surfaces in $\mathbb{R}^n$.

In $\mathbb{R}^2$ consider the curve on the unit quarter circle $x^2 + y^2 = 1$ in the first quadrant. This curve's equation is given by some parametrisation $x = \cos t, y = \sin t$ for $t \in [0, \pi/2]$. We denote this parametrisation by $P(t) = (\cos t, \sin t) = \cos t\ \hat{\imath} + \sin t\ \hat{\jmath}$.

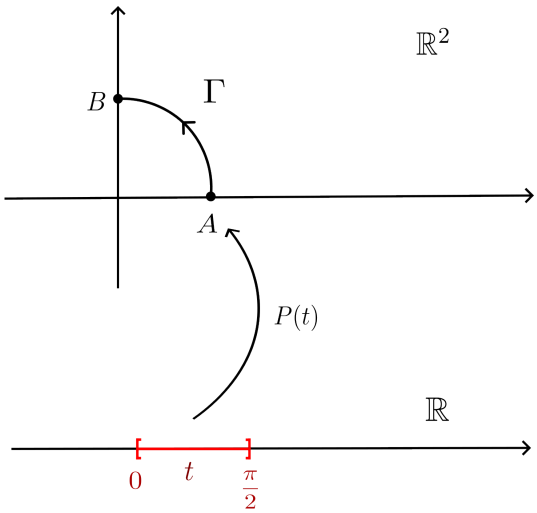


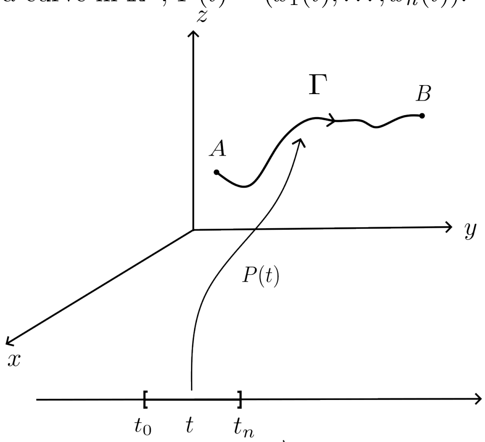
Figure 2.1.: $P(t) = (\cos t, \sin t)\,,\ t \in [0, \pi/2]$.

In Fig. 2.1 the curve's direction is anticlockwise due to the parametrisation $P(t)$ being defined from $t = 0$ to $t = \pi/2$. The curve would have clockwise direction if we took the parametrisation $-P(t)$ i.e. from $t = \pi/2$ to $t = 0$ instead. In Fig. 2.1, A is the initial point while B is the terminal point.

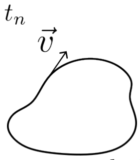
So we have the notion of clockwise and anticlockwise directions for curves in $\mathbb{R}^2$, but for curves in $\mathbb{R}^n$ we can't use this notion. For a curve in $\mathbb{R}^n$, $P(t) = (x_1(t), \ldots, x_n(t))$.

We first consider space curves in $\mathbb{R}^3$. The curve here is very simple, so determining the direction is as simple as choosing an interval to parametrise the curve over (here $t \in [t_0, t_n]$).

But what about the sides of a solid body in $\mathbb{R}^3$? How do we determine which side is positive or negative ?



In order to have a notion of direction in this case, we have to define a unit tangent vector $\vec{v}$ at each point on the curve.



We also want to exclude curves with corners (called *piecewise smooth directed curves*).

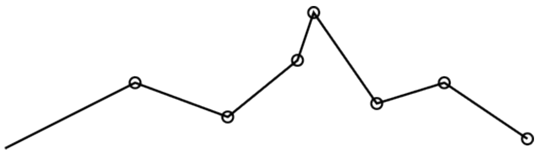


Figure 2.2.: We want to avoid curves with corners like the ones circled here.

Finally, as in the case of the side of a solid body, we want to include the case $A = B$ in which case we have a *closed curve.* This leads to cases where the parametrisation $P(t)$ may not be injective for all $t \in [a, b]$, as in Fig. 2.3. Although we allow closed curves, we want to

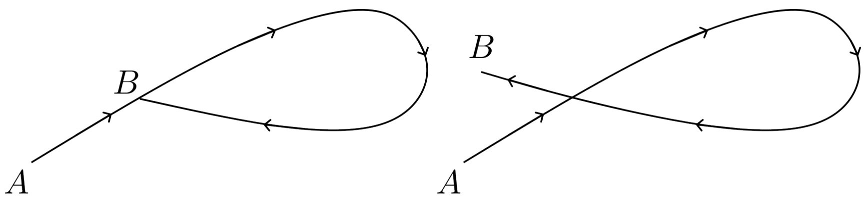


Figure 2.3.: We want to avoid curves that self-intersect.

avoid self-intersecting curves like Fig. 2.3. This is because one of our reasons for defining directed curves is to be able to do line integration along curves in $\mathbb{R}^n$, and developing such a theory for self-intersecting curves is much more complicated. We will deal with this in the next semester using contour integration, in the sixth semester course *Complex Analysis.* For now, we will want $P(t)$ to be injective on all of $[a,b]$ except maybe the endpoints in order to include closed curves, i.e. $P(t)$ should be injective on $[a,b)$ and $(a,b]$. So we define:

2.1.2. Definition. — A **directed (or oriented) curve** in $\mathbb{R}^n$ is a quadruple $\{\Gamma, A, B, \vec{v}\}$ where Γ is a set of points in $\mathbb{R}^n$, A and B are points of Γ called (respectively) the initial and final points of Γ, and $\vec{v}$ is a unit vector in $\mathbb{R}^n$ called the initial direction, for which there exists a mapping $P : [a,b] \to \mathbb{R}^n$ called the parametrisation of Γ such that the following conditions hold

(a) there exists an open interval I containing $[a,b]$ and a mapping from I into $\mathbb{R}^n$ which has derivatives of all orders and which coincides with P on $[a,b]$;

(b) $P([a,b]) = \Gamma, P(a) = A, P(b) = B$ and $P'(a) = \alpha\vec{v}$ for some $\alpha > 0$;[1]

(c) $P'(t) \neq 0$ for all $t \in [a,b]$;

(d) P is injective on $[a,b)$ and $(a,b]$.

Physically, one may think of $[a,b]$ as an interval of time and $P(t)$ as the position of a particle at time $t \in [a,b]$ as the particle moves along the curve Γ. Then $P'(t)$ is the *velocity* and $\|P'(t)\|$ is the *speed* of the particle. If $\|P'(t)\| = 1$ then the particle is moving at *unit speed*. So a parametrisation $P(t)$ of a curve Γ is said to be **unit speed** if $\|P'(t)\| = 1$.

Now the distance travelled by the particle is simply the integral of its speed with respect to time from $t = a$ to $t = b$, so the length of the curve $\ell(\Gamma)$ (say) is given by the same integral formula:

2.1.3. Definition. — The **length of the curve** Γ parametrised by $P(t)$, $t \in [a,b]$ is

$$\ell(\Gamma) = \int_a^b \|P'(t)\|\mathrm{d}t.$$

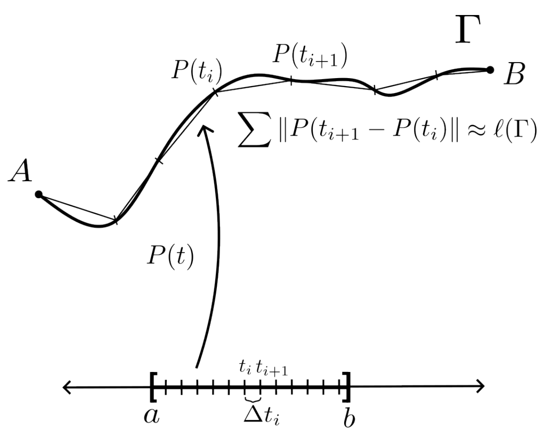


Figure 2.4.: Geometric intuition for $\ell(\Gamma)$

As P is differentiable, $P(t+\Delta t) = P(t) + P'(t)\Delta t + g(t, \Delta t)\,\Delta t$ for all t and $t+\Delta t \in [a,b]$ and as $\Delta t \to 0$, we have $P(t+\Delta t) - P(t) \to P'(t)\mathrm{d}t$ so

$$\sum \|P'(t_i)\|\Delta t_i \underset{\Delta t_i \to 0}{\longrightarrow} \int_a^b \|P'(t)\|dt = \ell(\Gamma).$$

[1] hence $\vec{v}$ is the *unit tangent vector*

2.2. Integration

Lecture 5
13^{th} Aug, 2025

2.2.1. Line integrals. —

Bibliography

[Apo74] Tom M. Apostol. *Mathematical Analysis*. Addison-Wesley, 1974.

[DK04a] Johannes J. Duistermaat and Johan A. C. Kolk. *Multidimensional Real Analysis I: Differentiation*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2004.

[DK04b] Johannes J. Duistermaat and Johan A. C. Kolk. *Multidimensional Real Analysis II: Integration*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2004.

[Rud53] Walter Rudin. *Principles of Mathematical Analysis*. McGraw Hill, 1953.